

미적 2-5

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1. $\nabla f(x, y, z) = \langle y+z, x+z, y+z \rangle$, $u = \langle \frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}, -\frac{1}{\sqrt{6}} \rangle$

$\langle 2, 4, 0 \rangle \cdot \langle \frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}, -\frac{1}{\sqrt{6}} \rangle = \frac{10}{\sqrt{6}}$

2. (1) $\nabla f(x, y) = \langle (y+1)\cos(xy+x), x\cos(xy+x) \rangle$ $|\langle 2, 0 \rangle| = 2$

$\nabla f(0, 1) = \langle 2, 0 \rangle$

(2) $\nabla f(x, y, z) = \langle \frac{yz}{1+x^2y^2}, \frac{xz}{1+x^2y^2}, \frac{xy}{1+x^2y^2} \rangle$ $|\langle \frac{1}{5}, \frac{1}{5}, \frac{2}{5} \rangle| = \frac{2\sqrt{5}}{5}$

3. (1) $\nabla f(x, y) = \langle ye^x, e^x \rangle$ $u = \langle \frac{2}{5}, -\frac{3}{5}, 2 \rangle = \langle -\frac{3}{5}, \frac{4}{5} \rangle$

$\langle 0, e^x \rangle \cdot \langle -\frac{3}{5}, \frac{4}{5} \rangle = \frac{4}{5}e^x$

(2) $\nabla f(2, 0) = \langle 0, e^2 \rangle = e^2$ 방향: $\langle 0, e^2 \rangle$, $2\sqrt{e^2} = 2e$

4. $\nabla F(x, y, z) = \langle y\cos(xy) - 2xy^2, x\cos(xy) - 2x^2y, xy\cos(xy) - 2xz \rangle$

$\eta = \nabla F(2, 1, 0) = \langle -4, -8, 2 \rangle$ $-4(x-2) - 8(y-1) + 2(z-0) = 0$ $2x+4y-2z=0$

5. $F(x, y, z) = x-y^2-z^2$ $\nabla F(x, y, z) = \langle 1, -2y, -2z \rangle$

$\eta = \nabla F(3, -1, -2) = \langle 1, 2, 4 \rangle$ $(x-3) - 2(y+1) + 2(z+2) = 0$ $x-2y+2z+1=0$

$x=3+t, y=-1+2t, z=-1+2t$

6. $\nabla f(x, y, z) = \langle 2x, 3y, -1 \rangle$ $\langle 2x, 3y, -1 \rangle = t \langle 1, 2, 3 \rangle$

$x = \frac{1}{2}t, y = \frac{1}{3}t, -1 = -t$ $x = -\frac{1}{6}, y = -\frac{1}{12}, t = -\frac{1}{3}$ $z = x^2 + 4y^2$

$z = \frac{1}{36} + \frac{1}{9} = \frac{1}{18}$ 접선의 방정: $(-\frac{1}{6}, -\frac{1}{12}, \frac{1}{18})$

7.

3-7

1. (1) $f_x(x, y) = 3x^2 - 6y$ $f_y(x, y) = -6x + 24y^2$ $y = \frac{1}{2}x$ $-6x + 24(\frac{1}{2}x)^2 = 0$

$x=0, 1$ $(0, 0), (1, \frac{1}{2})$ $f_{xx}(x, y) = 6x$ $f_{xy}(x, y) = -6$ $f_{yy}(x, y) = 48y$

$D(0, 0) = -36$ $D(1, \frac{1}{2}) = 108$ $D(0, 0) < 0$, $D(1, \frac{1}{2}) > 0$

$f_{xx}(1, \frac{1}{2}) = 6$ $f(1, \frac{1}{2}) = -1$

(2) $f_x(x, y) = 2x - y$ $f_y(x, y) = -x + y$ $(x, y) = (1, 2)$

$f_{xx}(x, y) = 2$ $f_{xy}(x, y) = -1$ $f_{yy}(x, y) = 0$ $D(1, 2) = -1$ $D(1, 2) < 0$

(3) $f_x(x, y) = 2xe^{\frac{y}{x}}$ $f_y(x, y) = e^{\frac{y}{x}} + \frac{1}{x}e^{\frac{y}{x}}(x+y) = \frac{1}{x}e^{\frac{y}{x}}(x+y+1)$ $(x, y) = (0, -2)$

$f_{xx}(x, y) = 2e^{\frac{y}{x}}$ $f_{xy}(x, y) = -\frac{y}{x^2}e^{\frac{y}{x}}$ $f_{yy}(x, y) = \frac{1}{x^2}e^{\frac{y}{x}}(x+y+2) = \frac{1}{x^2}e^{\frac{y}{x}}(x+y+1)$

$D(0, -2) = e^{-2}$ $f_{xx}(0, -2) = 2e^{-1}$ $D(0, -2) > 0$

2. $f_x(x, y) = f_y(x, y) = 0$ $(x, y) = (1, 1)$ $D(1, 1) = -6x(-1) - 36 = 12$

$f_{xx}(1, 1) < 0$ $(1, 1)$ 에서 $\nabla^2 f < 0$, f 는 $\nabla f = 0$ 에서 $\nabla^2 f < 0$ 이다.

3. $f_x(x, y) = 2x+2xy$ $f_y(x, y) = 2y+x^2$ $(x, y) = (0, 0)$, $(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$, $(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$

$x=1, -1$ $y=1$ $f(x, y) = 1+y^2y$ 3 $\frac{3}{4}$

$y=1, -1$ $x=1$ $f(x, y) = 2x^2+1$ 3 1 $3x+1, 3$ $3/5, 10$

$x=1, -1$ $y=1$ $f(x, y) = 1+y^2y$ 3 $\frac{3}{4}$

$y=1, -1$ $x=1$ $f(x, y) = 1$ 1 1

4.

13. 8

1. $\nabla f(x, y, z) = \lambda \nabla g(x, y, z) \rightarrow \langle 2x, 2y, 2z \rangle = \lambda \langle 1, 1, 1 \rangle$ $2x = \lambda, 2y = \lambda, 2z = \lambda, x+y+z = 12$

$x = \frac{\lambda}{2}, y = \frac{\lambda}{2}, z = \frac{\lambda}{2}$ $\lambda = 8$ $x=y=z=4$ $f(4, 4, 4)$

2. $\langle 4x, 2y, -2z \rangle = \lambda \langle 2x, \frac{3}{2}y, \frac{2}{3}z \rangle$ $4x = 2\lambda x, 2y = \frac{3}{2}\lambda y, -2z = \frac{2}{3}\lambda z, x^2 + y^2 + z^2 = 1$

$2x = \lambda, 2y = \lambda, -2z = \lambda$ $y = 0$ $\lambda = 4$ $2x = 0$ $\lambda = -4$ $y = 0$ $2x = 0$ $2y = 0$ $2z = 0$ $x = 0$ $y = 0$ $z = 0$ $(1, 0, 0)$ $(0, 1, 0)$ $(0, 0, 1)$ $(-1, 0, 0)$ $(0, -1, 0)$ $(0, 0, -1)$ $(1, 1, 1)$ $(-1, -1, -1)$ $(1, -1, -1)$ $(-1, 1, -1)$ $(1, 1, -1)$ $(-1, -1, 1)$ $(1, -1, 1)$ $(-1, 1, 1)$ $(1, 1, 1)$

6. 3.

4. $F(x, y, z) = x^2 y^2 + z^2$ $x^2 y^2 + z^2 = 3$ $\nabla F(x, y, z) = (2x, 2y, 2z) = t(-1, 1, 1)$

$\Rightarrow x = \frac{1}{2}t, y = \frac{1}{2}t, z = \frac{1}{2}t$ $\frac{3}{4}t^2 = 3$ $t^2 = 4$ $(x, y, z) = (-1, -1, -1), (1, 1, 1)$

(4.1)

(1): 1. $K = \{(x, y, z) \mid (x, y) \in P, 0 \leq z \leq \sqrt{4-x^2}\}$

(2)  $\iint_R \sqrt{4-x^2} dA = \iint_K 1 dV = \left(\frac{1}{2}\pi \times 2^2\right) \times 4 = 8\pi$

2. $\iint_D f_{xy}(x, y) dx dy = \int_0^1 \int_{-1}^1 f_{xy}(x, y) dx dy = \int_0^1 \int_{-1}^1 \left(\frac{\partial}{\partial x} f_y(x, y)\right) dx dy$

(1) $= \int_0^1 [f_y(x, y)]_0^1 dy = \int_0^1 (f_y(1, y) - f_y(0, y)) dy = \int_0^1 \frac{\partial}{\partial y} (f(1, y) - f(0, y)) dy$

(2) $= [f(1, y) - f(0, y)]_0^1 = (f(1, 1) - f(0, 1)) - (f(1, 0) - f(0, 0)) = 4 - 1 = 3$

3.

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